

18.1: Motion in 2 or 3 dimensions:

Constant Acceleration Models in 2D

Same SUVAT equations as 1D, just use vector quantities for each variable $\begin{pmatrix} n_x \\ n_y \end{pmatrix}$

Can use column vectors, or i, j notation

Acceleration is shown as $\begin{pmatrix} 0 \\ g \end{pmatrix} ms^{-2}$ downwards.

$$\text{Speed} = |\text{velocity}| \Rightarrow \vec{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix} \Rightarrow |v| = \sqrt{v_x^2 + v_y^2}$$

To find the Cartesian Equation of a point where $\vec{r} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} u(t) \\ v(t) \end{pmatrix}$

Write the 2 equations: $x = u(t)$ & $y = v(t)$

Rearrange for t, & substitute

For Calculus:

$$\vec{r} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \dot{r} = v = \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} \Rightarrow \ddot{r} = \dot{v} = a = \begin{pmatrix} \ddot{x} \\ \ddot{y} \end{pmatrix}$$

where: $\dot{n} = \dfrac{d(n)}{dt}$

Force as a function of Time:

$$\begin{pmatrix} F_1 \\ \vdots \\ F_n \end{pmatrix} = m \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$$

hence motion can be inspected.